UNIT I – Metal Cutting

23A03501 MACHINING PROCESSES / B.Tech Mech (JNTUK) – Exam Notes

I. Elementary Theory of Metal Cutting

Metal cutting = removing unwanted material from a workpiece in the form of **chips** using a wedge-shaped, harder tool, to get required shape, size & finish.

Elements of cutting process

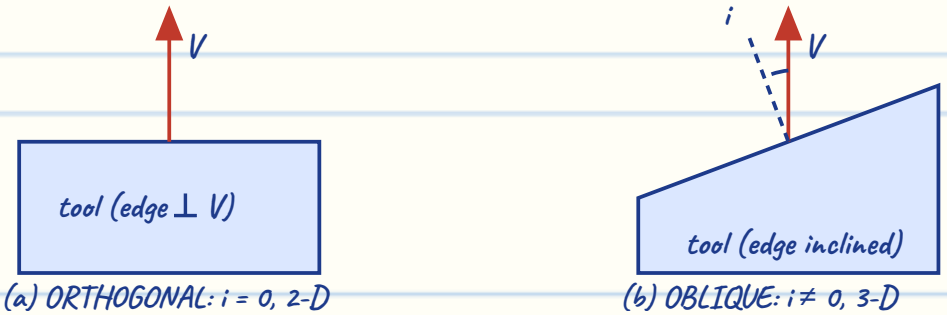
- **Workpiece** (softer) + **cutting tool** (harder wedge) + **machine**
- Relative motion tool/work: **primary (cutting) motion** → speed V
- **Secondary (feed) motion** → feed f; **depth of cut** d (infeed)
- Result: **chip** (by shearing in **shear zone**) + machined surface

Cutting parameters (turning)

- **Cutting speed** $V = \pi DN/1000$ m/min (D mm, N rpm)
- **Feed** f = mm/rev (lathe), mm/stroke (shaper), mm/tooth (milling)
- **Depth of cut** $d = (D_1 - D_2)/2$ mm, ⊥ to feed
- Uncut chip thickness $t = f \sin(\text{SCEA-complement})$; width $b = d / \sin\varphi_s$

$$MRR = 1000 \cdot V \cdot f \cdot d \ \ mm^3/min \quad | \quad \text{Machining time } Tm = L / (f \cdot N) \ min$$

Orthogonal vs Oblique cutting



(a) ORTHOGONAL: $i = 0$, 2-D (b) OBLIQUE: $i \neq 0$, 3-D

Fig 1. Orthogonal and oblique cutting (plan view); i = inclination angle

Orthogonal	Oblique
Edge ⊥ cutting velocity (i = 0)	Edge inclined at i to the normal
2 force components (Fc, Ft)	3 components (Fc, Ft, Fr)
Chip flows ⊥ edge, plane strain	Chip flows sideways, curls helically
Edge longer than chip width	Longer edge contact → better life, finish
Used for theory; e.g. parting, broaching	Practical turning, milling, drilling

2. Single Point Cutting Tool – Geometry

- **Shank** – held in tool post; **Flank** – surface below edge facing machined surface
- **Face (rake face)** – chip slides over it; **Heel** – lowest portion of tool
- **Nose** – junction of side & end cutting edges (radius r)
- **Side cutting edge** does main cutting; **end cutting edge** clears finished surface

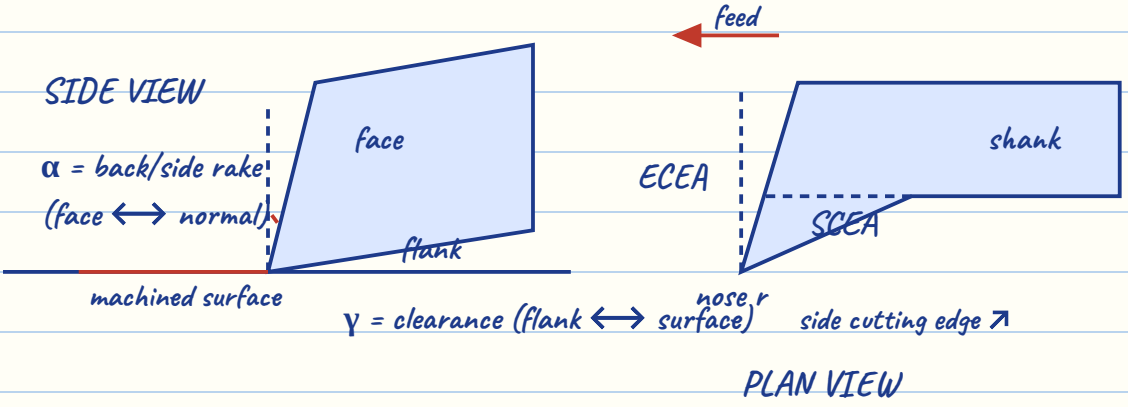


Fig 2. Single point tool – side and plan views with angles

Tool angles & functions

- **Back rake** αb (along shank), **side rake** αs (across) – chip flow, force
  - +ve rake: ↓ force, ↓ power, better finish, but weak edge (ductile, HSS)
  - –ve rake: strong edge, takes shocks (carbide/ceramic, hard work)
- **Relief (clearance)** end & side (5–10°): avoids rubbing on work
- **SCEA** Cs (15–30°): ↑ edge length, thins chip; **ECEA** Ce (5–15°): finish
- **Inclination angle** λ: directs chip (–ve λ → chip away from finished surface)
- **Nose radius** r: ↑ strength & finish; too big → chatter

Tool signature (designation)

System	Order of angles	Example
ASA (American, tool-in-hand)	αb – αs – end relief – side relief – ECEA – SCEA – r	8-14-6-6-6-15-1 (mm)
ORS (orthogonal rake, tool-in-use)	λ – αn – side relief γ – end relief γ_1 – ECEA φ_1 – cutting edge angle φ – r	10-8-6-6-6-75-1

- ORS principal cutting-edge (approach) angle $\varphi = 90° - SCEA(ASA)$
- Mnemonic ASA: **"Back Side End Side End Side Radius"** (3 rake/relief pairs)

3. Chip Formation

- Tool pushes into work → **compressive stress** → plastic deformation by **shear** along the **shear plane** (angle φ)
- Chip = material sheared, then slides on rake face (friction: **secondary deformation**)
- Chip is thicker ($t_c > t$) & shorter than uncut layer → **chip compression**

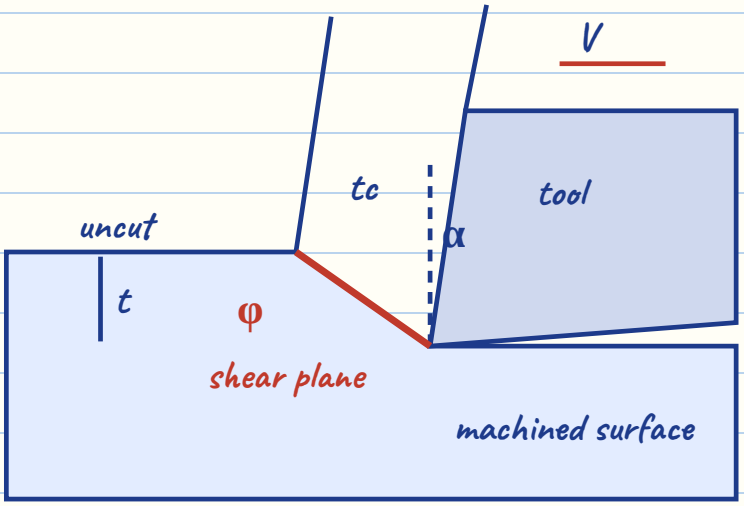


Fig 3. Orthogonal cutting model – shear plane, t, tc, φ, α

Chip thickness ratio & shear angle

$$r = t / t_c = \sin\varphi / \cos(\varphi - \alpha) \quad (r < 1)$$

$$\tan\varphi = r\cos\alpha / (1 - r\sin\alpha)$$

- Chip velocity $V_c = r \cdot V$; shear velocity $V_s = V\cos\alpha / \cos(\varphi - \alpha)$
- Shear strain $\gamma = \cot\varphi + \tan(\varphi - \alpha)$; also $r = V_c/V = L/L_c$ (length ratio)
- Large φ → thinner chip, less deformation, less force & heat (good)

★ Important

- Derivation: $AB = t/\sin\varphi = tc/\cos(\varphi-\alpha) \Rightarrow r = \sin\varphi / \cos(\varphi-\alpha)$
- Expand $\cos(\varphi-\alpha) \rightarrow r\cos\alpha + r\sin\alpha\tan\varphi = \tan\varphi \Rightarrow \tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$

4. Types of Chips

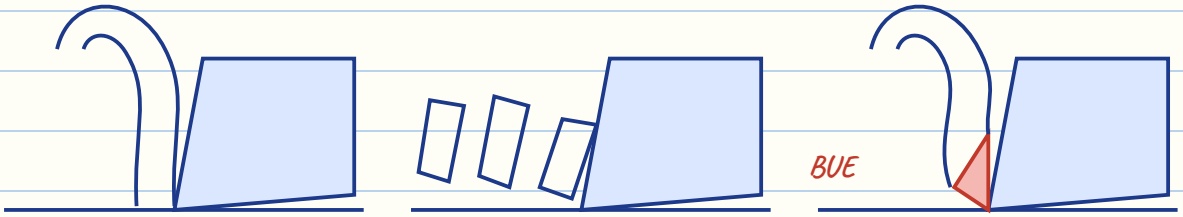


Fig 4. Types of chips

Type	Occurs when	Remarks
Continuous (ribbon)	Ductile metal, high V, small f, +ve rake, good coolant	Best finish, steady force; long chips = hazard → use chip breaker
Discontinuous (segmented)	Brittle (CI, bronze), low V, large f & d, −ve rake	Fluctuating force, rough finish, easy to handle
Continuous with BUE	Ductile, low–medium V, high friction	Poor finish, unstable edge, tool wear

- Also **serrated (shear-localised)** chips – Ti alloys, hard steels at high V

5. Built-Up Edge (BUE)

BUE = layer of work material **welded (pressure-weld) to the tool face** near the tip, strain-hardened, acting as a false cutting edge.

- **Causes:** high friction + high pressure + temperature, ductile work, low–medium V, no coolant/lubricant, small +ve rake, affinity of metals
- **Process:** builds up → becomes unstable → breaks off → carried by chip & work surface → repeats

Effects

- (−) rough, torn surface; poor dimensional accuracy (changes tool geometry)
- (−) tool crater/flank wear by fragments; force fluctuation, vibration
- (+) protects the tool edge, ↑ effective rake → ↓ force (at low speed, roughing only)

Remedies / how to eliminate

- ↑ cutting speed (above ~ recrystallisation temperature) · ↓ feed
- ↑ +ve rake angle · use sharp, polished tool; lower friction
- Effective **cutting fluid** (EP additive) · tool with low affinity (carbide/coated)

6. Chip Breakers

- Long continuous chips → danger to operator, tangle, damage surface, poor coolant access
- **Chip breaker** curls & fractures chip into short pieces (6-shaped / 'C' chips)

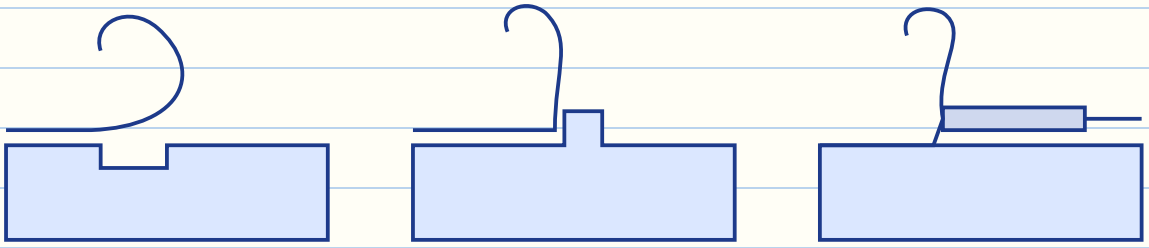


Fig 5. Chip breakers

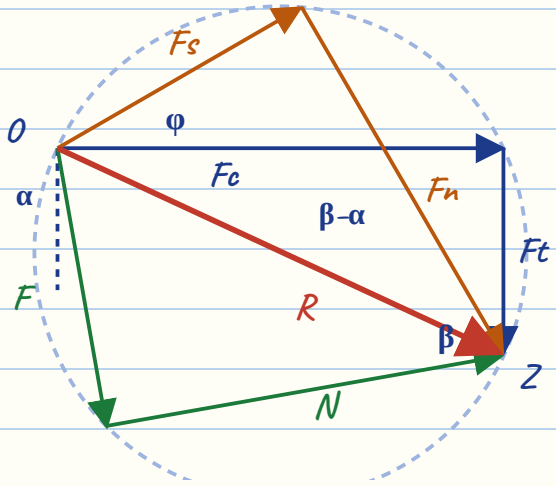
- **Groove type** – ground groove behind edge (width, depth, radius) – chip curls in groove
- **Obstruction type** – raised step/ridge ahead of edge deflects chip → bends → breaks
- **Clamp-on (plate) type** – adjustable carbide plate clamped on insert; common on indexable inserts
- Also by feed/speed variation, or **moulded insert geometry**; good break = f↑, d↑

7. Mechanics of Orthogonal Cutting – Merchant's Analysis

Assumptions

- Sharp tool, no rubbing on flank; 2-D (plane strain) orthogonal cutting
- Continuous chip, no BUE; shear takes place on single thin plane; uniform stress on it
- Constant V, constant t; tool rigid; chip in equilibrium under two equal-opposite forces

Merchant's force (circle) diagram



R = resultant (diameter)
Fc cutting force (power)
Ft thrust force (⊥ V)
F friction on rake face
N normal to rake face
Fs shear force (on plane)
Fn normal to shear plane
α rake, φ shear, β friction angle

Fig 6. Merchant's circle (R as diameter; each force pair makes a right angle on circle)

- Tool exerts R on chip (= R' by chip on tool): in equilibrium → R splits 3 ways
- (Fc, Ft) measured by dynamometer; (F, N) on rake face; (Fs, Fn) on shear plane
- Friction: **$\mu = F/N = \tan\beta$**, β = friction angle; $Ft/Fc = \tan(\beta - \alpha)$

Derived equations (from circle geometry)

$$F = Fc \sin\alpha + Ft \cos\alpha \qquad N = Fc \cos\alpha - Ft \sin\alpha$$

$$Fs = Fc \cos\varphi - Ft \sin\varphi \qquad Fn = Fc \sin\varphi + Ft \cos\varphi$$

$$\mu = F/N = (Ft + Fc \tan\alpha)/(Fc - Ft \tan\alpha) \;;\; \beta = \tan^{-1}\mu$$

$$R = Fc / \cos(\beta - \alpha) \;;\; Fs = R \cos(\varphi + \beta - \alpha)$$

- Shear stress $\tau s = Fs / As$, with $As = b \cdot t / \sin\varphi$ (shear plane area)
- Also $Fn = Fs \tan(\varphi + \beta - \alpha)$; $Fc = Fs \cos(\beta-\alpha)/\cos(\varphi+\beta-\alpha)$

Power & specific energy

$$Pc = Fc \cdot V \quad (W,\ V \text{ in } m/s) \;;\; u = Pc / MRR = Fc / (b \cdot t)$$

- Total energy = shear energy (Fs·Vs) + friction energy (F·Vc) → ~ 70–80 % in shear
- u (specific cutting energy): steel ~2–5, Al ~0.7–1, CI ~1–2 J/mm^3
- Size effect: u ↑ when t ↓ (finishing cuts need more energy per volume)
- Machine motor power = Pc / η ($\eta \approx 0.7$–0.85)

Merchant's shear angle relations

- **1st theory** – shear plane is where **energy (cutting force) is minimum**; τs constant: put $dFc/d\varphi = 0$

$$\varphi = 45° + \alpha/2 - \beta/2$$

- **2nd theory** – τ_s varies linearly with normal stress σ ($\tau_s = \tau_0 + k\sigma$):

$$2\varphi + \beta - \alpha = C \quad (C = \text{machining constant of material})$$

- Lee & Shaffer (slip-line): $\varphi = 45^\circ + \alpha - \beta$; Ernst–Merchant is the 1st form
- Meaning: ↑ rake α or ↓ friction β ⇒ ↑ φ ⇒ ↓ chip thickness, ↓ force (agrees with practice)
- Experimental values deviate for real metals – hence 2nd theory/C

★ *Important*

- Remember: $\varphi = 45 + \alpha/2 - \beta/2$; $2\varphi + \beta - \alpha = C$; $\mu = \tan\beta$
- $F_c \propto$ power; $F_t \propto$ deflection; $F/N = \mu$; $r = \sin\varphi/\cos(\varphi-\alpha)$

Solved Example 1 (Merchant's circle)

In orthogonal cutting: $t = 0.25$ mm, $t_c = 0.75$ mm, $\alpha = 10^\circ$, $F_c = 900$ N, $F_t = 450$ N, $V = 60$ m/min. Find φ, F_s, F_n, F, N, μ, power.

1. $r = 0.25/0.75 = 0.333$; $\tan\varphi = 0.333\cos10^\circ/(1 - 0.333\sin10^\circ) = 0.3485$ ⇒ **$\varphi = 19.2^\circ$**
2. $F_s = F_c\cos\varphi - F_t\sin\varphi = 900(0.944) - 450(0.329) =$ **702 N**
3. $F_n = F_c\sin\varphi + F_t\cos\varphi = 296 + 425 =$ **721 N**
4. $F = F_c\sin\alpha + F_t\cos\alpha = 156 + 443 =$ **600 N** ; $N = F_c\cos\alpha - F_t\sin\alpha = 886 - 78 =$ **808 N**
5. $\mu = F/N = 0.742 \Rightarrow \beta = 36.6^\circ$ (check: $F_t = F_c\tan(\beta-\alpha) = 900\tan26.6^\circ = 450$ ✓)
6. Power $P_c = F_c \cdot V = 900 \times 1$ m/s = **900 W**

8. Heat Generation in Metal Cutting

- Nearly all cutting work → **heat**; raises temperature of tool, chip, work

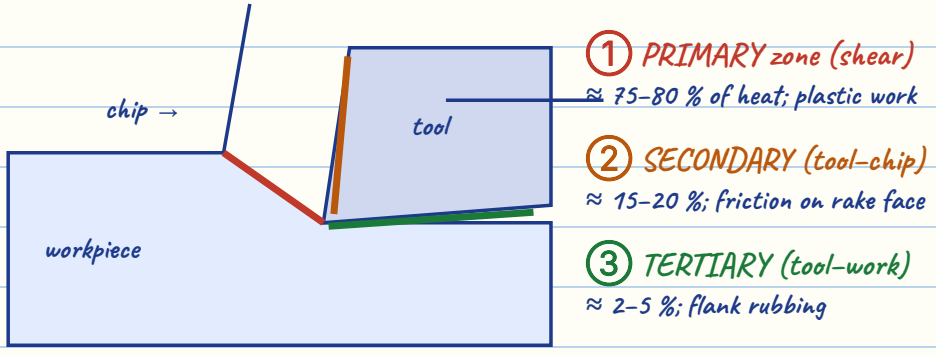


Fig 7. Heat zones in orthogonal cutting

- **Heat carried away**: chip ≈ 75–80 % · work ≈ 10–15 % · tool ≈ 5–10 % (at normal V)

- Max temperature on rake face ≈ at a distance behind edge (end of tool–chip contact)
- Temp. ↑ with V (strongest), f, d (weakly); $\theta \propto V^{0.4}$ (approx.)
- Effects: thermal softening of tool → **wear, BUE, crater**; work thermal expansion → inaccuracy; residual stress
- Control: coolant, sharp +ve rake tool, lower V, tool with low conductivity loss (carbide)
- Temperature measured by: tool–work thermocouple, embedded thermocouple, IR camera, calorimetry

9. Tool Wear & Tool Life

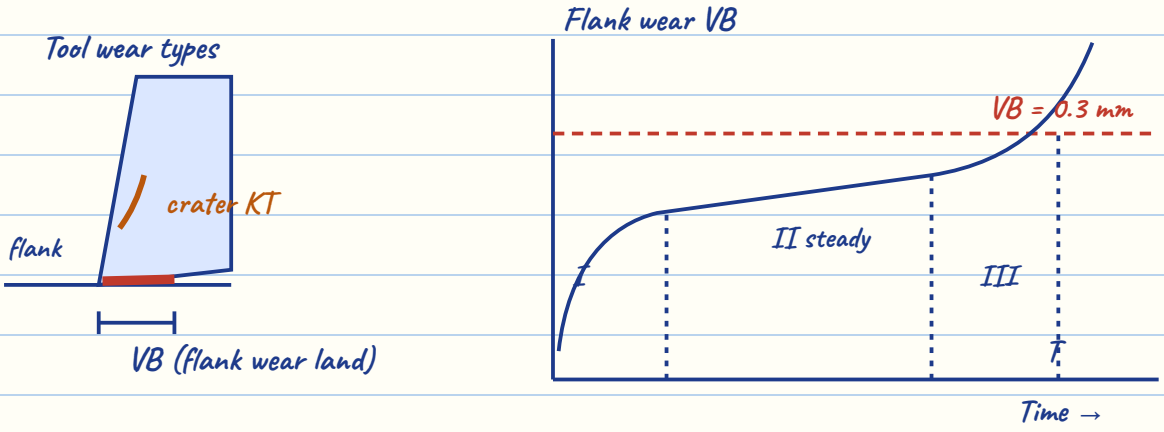


Fig 8. Wear types (flank, crater) and wear–time curve

- **Flank wear** (VB) – on clearance face, by abrasion; ↑ forces, ↓ accuracy/finish
- **Crater wear** (KT, KM) – on rake face, by diffusion/adhesion at high temp.
- Other: nose wear, notch (DOC-line) wear, chipping, edge fracture, plastic deformation
- **Mechanisms**: abrasion, adhesion (attrition), diffusion, oxidation, fatigue, thermal cracking
- Stages: **I** initial rapid (edge rounding) · **II** steady (long) · **III** rapid failure

Tool life criteria (T = time of cutting until failure)

- **Flank wear VB = 0.3 mm** (uniform) or 0.6 mm (non-uniform) – ISO 3685; HSS ~0.75–1.5 rough
- Crater depth KT = 0.06 + 0.3 f (mm) · complete failure · surface finish deterioration
- ↑ cutting force / power / temperature · dimensional change · chatter or sound

Taylor's tool life equation

$$V \cdot T^{n} = C$$

- V = cutting speed (m/min), T = tool life (min), n = tool-life exponent, C = constant (speed for $T = 1$ min)
- n depends on tool material (see table); $\log V = \log C - n \log T$ → **straight line**

Tool material	n
HSS	0.08 – 0.20 (≈ 0.125)
Cemented carbide	0.20 – 0.40 (≈ 0.25)
Ceramic	0.40 – 0.70 (≈ 0.5)

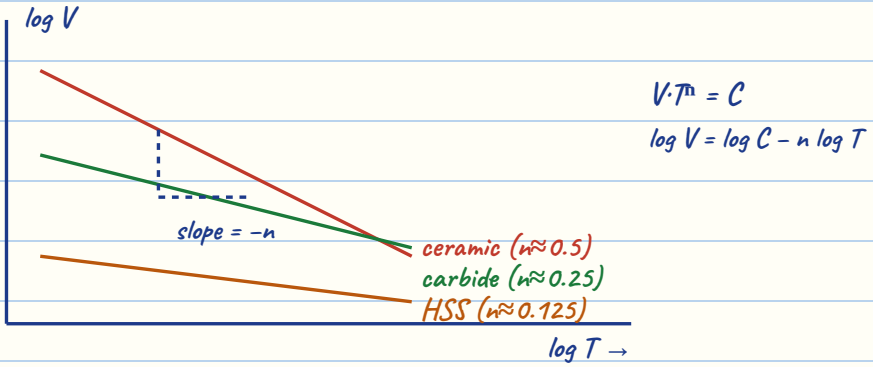


Fig 9. Tool life curves on log–log scale

Extended Taylor equation & factors affecting tool life

$$V \cdot T^{n} \cdot f^{a} \cdot d^{b} = C \quad (a \approx 0.15\text{–}0.4,\ b \approx 0.1\text{–}0.2,\ n \approx 0.1\text{–}0.5)$$

- V has the **greatest** effect, then f, then d ($V \gg f > d$)
- Also: tool material/geometry, work hardness & microstructure, coolant, rigidity of machine, BUE
- Taylor ratio: $(V_1/V_2) = (T_2/T_1)^{n}$; 20 % ↑ in V roughly halves life of HSS tool

Solved Example 2 (Taylor)

Tool life 60 min at $V = 30$ m/min and 20 min at $V = 45$ m/min. Find n, C and life at $V = 36$ m/min.

1. $V_1 T_1^{n} = V_2 T_2^{n} \Rightarrow (45/30) = (60/20)^{n} \Rightarrow n = \ln 1.5 / \ln 3 =$ **0.369**
2. $C = 30 \times 60^{0.369} = 30 \times 4.53 =$ **136**
3. $T = (C/V)^{1/n} = (136/36)^{2.71} = (3.78)^{2.71} \approx$ **36.6 min**

10. Cutting Fluids (Coolants)

Functions

- **Cool** tool & work (↓ temp) · **Lubricate** chip–tool interface (↓ friction, ↓ BUE)
- **Wash away chips** (swarf) · **protect** work/machine from rust
- ↑ tool life, ↑ surface finish, ↑ allowable V, ↓ power, ↑ dimensional accuracy

Types

Type	Composition	Main use
Straight (neat) oils	Mineral/fatty oils + EP additives (S, Cl, P)	Lubrication: low V, tough operations – threading, broaching
Soluble oils (emulsions)	Oil droplets in water 1:20–1:50 + emulsifier	Cooling + some lubrication: general turning, milling; cheap
Semi-synthetic	Small oil + chemicals in water	Mixed duty, grinding
Synthetic	Chemical (no oil) in water	High speed, grinding, best cooling, clean
Gases / solids	Compressed air, CO_2 ; graphite, MoS_2	Dry/near-dry, cast iron; high-temp lubrication

- Additives: **EP** (extreme-pressure) – sulphur, chlorine, phosphorus; anti-foam, biocide, corrosion inhibitor

Selection & application

- High V (low load) → need **cooling** → water-based; low V (high load) → **lubrication** → oil
- Steel: soluble / sulphurised oil · Al: kerosene/soluble oil · CI, brass: dry or air jet
- Mg: **never water-based (fire)** · carbide tools: dry or copious flood (avoid thermal shock)
- Methods: **flood**, mist/spray, manual, through-tool (high-pressure), minimum quantity lubrication (MQL)
- Disadvantages: cost, disposal, health (dermatitis, mist), bacteria – hence dry/MQL machining

II. Machinability

Machinability = ease with which a material can be machined, judged by **tool life, cutting force/power, surface finish, chip form.**

- Factors: work – hardness, strength, ductility, microstructure, thermal conductivity, work-hardening, composition (S, Pb, P improve; Cr, Ni, Mo reduce)
- Process – speed, feed, tool material/geometry, rigidity, cutting fluid

$$\text{Machinability index (\%)} = V_{60} \text{ of material} / V_{60} \text{ of free-cutting steel (B1112)} \times 100$$

- V_{60} = cutting speed giving 60 min tool life; free-cutting steel B1112 = **100 %**
- Typical: Al alloys 300–1500 · free-cutting brass ~ 200 · mild steel ~ 55–70 · grey CI ~ 60–80 · stainless steel ~ 25–45 · Ti alloys ~ 20

- Improve: free-machining additives (S, Pb – **resulphurised steels**), proper heat treatment (annealing), cold drawing, right tool & coolant

12. Economics of Machining

- Aim: choose **V** (and f, d) for **min. cost per piece** or **max. production rate**
- Low V → big machining time; high V → rapid tool wear → many tool changes ⇒ **an optimum V**

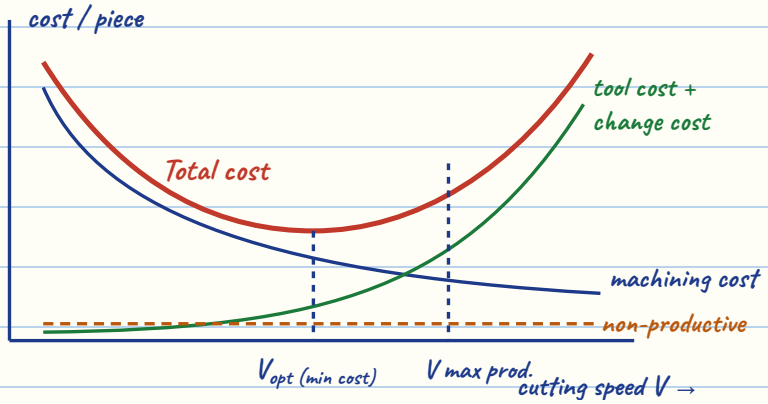


Fig 10. Cost components vs cutting speed

Cost per piece

$$C_p = C_o \cdot t_l + C_o \cdot t_m + (C_o \cdot t_c + C_t) \cdot (t_m / T)$$

- C_o = machine + operator rate (₹/min); t_l = loading/unloading time; t_m = machining time/piece
- t_c = tool change time (min); C_t = tool cost per edge (incl. regrind); t_m/T = tool changes per piece
- $T_m = \pi D L/(1000 \cdot f \cdot V)$ and $V = C/T^n$, so $t_m \propto 1/V$

Optimum tool life & speed

$$T_{min\ cost} = (1/n - 1)(t_c + C_t/C_o)$$

$$T_{max\ production} = (1/n - 1) \cdot t_c$$

- Then $V_{opt} = C / T_{opt}^{\,n}$; production time/piece = $t_l + t_m + t_c \cdot t_m/T$
- **V(max production) > V(min cost)** (since $C_t/C_o > 0$); between = **high-efficiency range**
- Min-cost used normally; max-production in emergency/bottleneck
- Higher n (ceramic) ⇒ lower optimum T ; Cost ↑ if t_c, C_t large ⇒ run slower

13. Cutting Tool Materials

Requirements

- High **hot hardness** (red hardness) · wear/abrasion resistance · toughness & shock resistance
- Low friction/affinity to work · chemical stability · thermal conductivity & low expansion · cheap, easily shaped

Material	Composition	Hot hardness limit	Characteristics / uses
HCS (high carbon steel)	0.8–1.3 % C	≈ 250 °C	Cheap, V ≈ 10 m/min; hand tools, taps, files
HSS	18W-4Cr-1V (T1); Mo type M2; Co added (T15)	≈ 600 °C	Tough, V ≈ 30–50; drills, reamers, milling cutters, complex tools
Cast alloys (Stellite)	Co 40-50, Cr 30, W 15–20	≈ 800 °C	Cast & ground, not heat-treated; V ≈ 1.5–2 × HSS; brittle
Cemented carbide	WC grains + Co binder (sintered); + TiC, TaC for steel	≈ 1000 °C	V ≈ 100–200; hard, brittle; indexable inserts; K (CI), P (steel), M grades
Coated carbide	TiN, TiC, Al_2O_3 layers (CVD/PVD) 3–10 µm	> 1000 °C	↑ wear & crater resistance, ↓ friction; 2–3 × life
Ceramics	Al_2O_3 (+ ZrO_2/TiC), Si_3N_4	≈ 1200 °C	Very hard, chemically inert, V ≈ 200–600; brittle – CI, hardened steel, rigid set-up
Cermets	TiC/TiN + Ni–Mo binder	≈ 1100 °C	Between carbide & ceramic; finishing steels, good finish
CBN	Cubic boron nitride, 2nd hardest	≈ 1400 °C	Hardened steels (> 50 HRC), superalloys; no reaction with Fe
Diamond (PCD/single)	Polycrystalline diamond compact	≈ 700 °C (in air)	Hardest; Al, Cu, plastics, composites; **not for ferrous** (carbon diffuses)

- Trend of hardness ↑ : HCS < HSS < Cast alloy < Carbide < Ceramic < CBN < Diamond (toughness ↓)
- Mnemonic: **"Hot Hardness Climbs: C-S-C-C-C-C-D"**

14. Quick Revision & Likely Exam Questions

★ Important

- $r = t/tc = \sin\varphi/\cos(\varphi-\alpha)$; $\tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$
- $F = Fc\sin\alpha + Ft\cos\alpha$; $N = Fc\cos\alpha - Ft\sin\alpha$; $\mu = F/N = \tan\beta$
- $Fs = Fc\cos\varphi - Ft\sin\varphi$; $Fn = Fc\sin\varphi + Ft\cos\varphi$; $Pc = Fc\cdot V$
- $\varphi = 45 + \alpha/2 - \beta/2$; $2\varphi + \beta - \alpha = C$
- $VT^n = C$; n: HSS 0.125, carbide 0.25, ceramic 0.5 ; VB = 0.3 mm
- $T(\text{min cost}) = (1/n - 1)(tc + Ct/Co)$; $T(\text{max prod.}) = (1/n - 1)\,tc$
- Heat: shear 75–80 %, friction 15–20 %, flank 2–5 %

Likely questions

1. Differentiate orthogonal & oblique cutting with neat sketches. (5M)
2. Draw a single-point tool; explain tool angles; write ASA & ORS signature. (10M)
3. Types of chips with sketches and conditions of formation. (5/10M)
4. What is BUE? Causes, effects, remedies. (5M)
5. Chip breakers – explain types with diagrams. (5M)
6. Derive relation of shear angle and chip thickness ratio. (5M)
7. Draw Merchant's circle; derive Fs, Fn, F, N in terms of Fc, Ft. (10M)
8. Merchant's two shear-angle theories; numerical on circle. (10M)
9. Heat zones and heat distribution in cutting. (5M)
10. Tool wear types, tool life criteria; Taylor's equation + numerical. (10M)
11. Cutting fluids – functions, types, selection. (5/10M)
12. Machinability & machinability index; factors. (5M)
13. Economics: derive optimum speed for min cost / max production. (10M)
14. Cutting tool materials with properties & applications. (10M)